A

B

WT 2.599 Å

A59P
G25E
G25E-K30H
G27D
G27D-K30H
G27D-K30H-Q58M
G27T
G27T-K30H
G27T-K30H-Q58M
K30H
K30H-Q58M
K60R
M132D
M132E
Q58M
R28A
R28S
T31I
T31L
T31V
Y32F
E72A
G27D-M130G
N57A
N57W
T31A
T31D
WT–GCG
WT–GCA

Distance unavailable

0
2
4
6

Target C4N4–E72 distance (Å)

0
20
40
60

Deviation from WT (SD)

Active
Inactive
Partially inactive
Not tested
Designed inactive control (not measured)
Wild-type reference
Internal CP (long-range)
Pocket CP
Protein–target-C CP
±3 SD band